



QE Data Software

User Guide & Training Tutorial

A standalone, zero-install workbench for CMM inspection data,
PC-DMIS routine auditing, SPC capability, and 8D root-cause reporting.

Part Master

Routine Auditor

Data Ingestion

SPC & Capability

8D / 5-Why

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Getting Started

QE Data Software is a standalone, browser-based workbench for quality engineers and CMM programmers. It runs entirely on your own machine — there is no server, no installer, and no account to create. Everything you enter is stored locally in your browser and stays on your computer unless you export it.

Opening the App

- 1 Unzip the QE Data Software folder**
Extract it anywhere on your computer — your Desktop, a network share, or a USB drive all work.
- 2 Double-click index.html**
It opens in your default web browser (Chrome, Edge, or Firefox all work well). No installation or admin rights are required.
- 3 Start working**
The Dashboard loads immediately. Use the five tabs across the top to move between modules at any time — your place is never lost, since each module keeps its own data.

TIP

The app needs an internet connection the first time it loads a page, only to fetch its Tailwind CSS styling from a CDN. The SPC charting engine (Chart.js) is bundled inside the app folder, so charts and all data entry work fully offline once the page has loaded.

How Your Data Is Stored

All part records, inspection runs, routine audits, and 8D investigations are saved in your browser's local database (IndexedDB) — not on any server. That means:

- Your data persists automatically between sessions, as long as you reopen the app in the **same browser on the same computer**.
- Nothing is uploaded anywhere. This makes the tool safe to use on a shop-floor machine with no internet access.
- Clearing your browser's site data/history for this page will erase your records — use the Backup Data button regularly (see Chapter 7).
- To move data to another computer, export a backup file and restore it there.

The Five Modules

Tab	What it's for
Dashboard	Manage part numbers, revisions, and characteristic tolerances; see summary stats.
Routine Auditor	Paste PC-DMIS program text to catch alignment and vector logic risks.
Data Ingestion	Import CSV/TXT inspection results and flag out-of-spec characteristics.
SPC & Capability	Trend charts and Cp/Cpk for any characteristic across all its runs.
8D / 5-Why	Build and print formal root-cause investigations from any flagged anomaly.

Dashboard & Part / Program Master

The Dashboard is your home base and your master list of parts. Every characteristic tolerance you define here is what the Data Ingestion and SPC modules compare measurements against, so set this up before importing inspection data.

Creating a Part Record

- 1 **Click “+ New Part Record”**
Opens the part editor. Enter the Part Number (required), Revision Level, CMM Program ID, and a short description.
- 2 **Add characteristics**
Click “+ Add Characteristic” for each dimension you inspect. Enter a Label (must match what your CSV/PC-DMIS export calls it), Nominal, Tol +, Tol -, Unit, and Type (Linear +/-, GD&T, or Basic/reference-only).
- 3 **Save**
Click “Save Part Record.” The part now appears in the master list and is available to select in Data Ingestion, SPC, and 8D.

QE Data Software
CMM Metrology & Quality Engineering Workbench

Dashboard Routine Auditor Data Ingestion SPC & Capability 8D / 5-Why

PART RECORDS 0 INSPECTION RUNS 0 OUT-OF-SPEC RATE — OPEN 8D INVESTIGATIONS 0

Part & Program Master

PART NUMBER

Recent Inspection Runs

TIMESTAMP

New Part Record

Part Number * 10245-B Revision Level C

CMM Program ID 10245B_FINAL-PRG Description Machined bracket, aluminum 6061-T6

Characteristics (Nominals & Tolerances) + Add Characteristic

LABEL	NOMINAL	TOL +	TOL -	UNIT	TYPE
DIA_1	25.000	0.020	0.020	mm	Linear +
LENGTH_A	104.500	0.050	0.050	mm	Linear +
POS_TRUE	0.000	0.250	0.000	mm	GD&T

Cancel Save Part Record

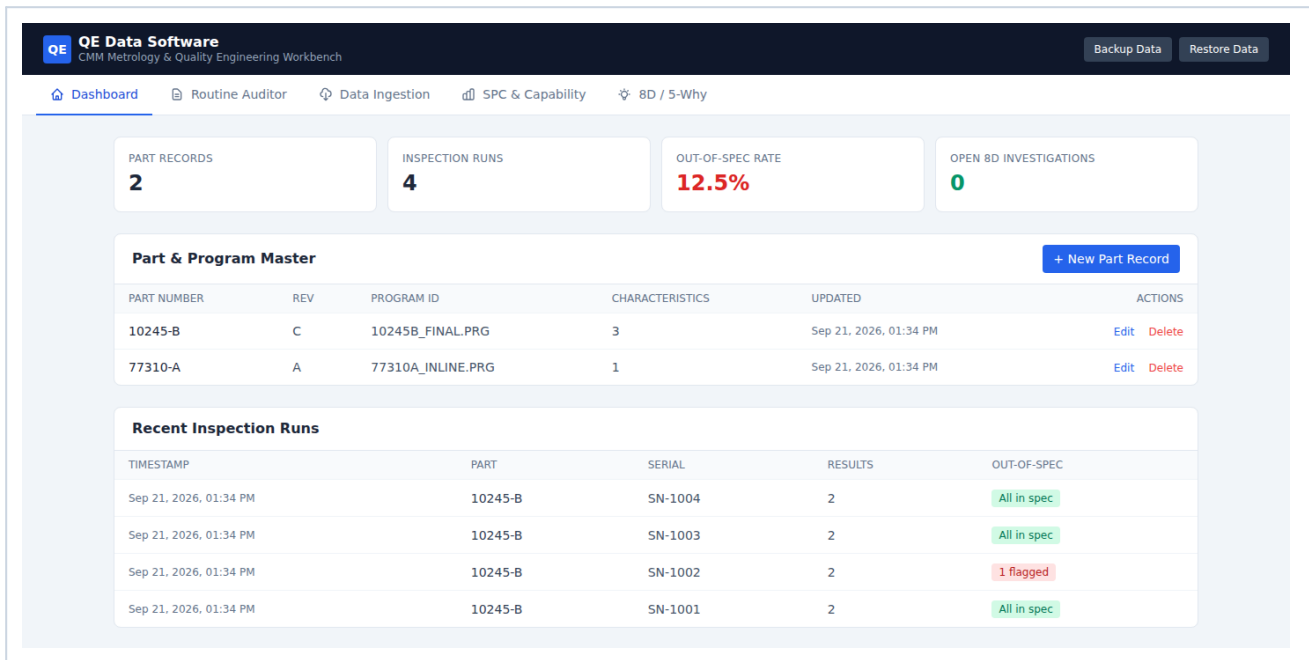
The Part Record editor — add one row per characteristic with its nominal and tolerance band.

IMPORTANT

The Label you type here (e.g. “DIA_1”) must match the characteristic name in your inspection CSV exactly (case-insensitive). That’s how the app links an imported measurement back to its tolerance.

Reading the Dashboard

Once you have parts and runs, the Dashboard summarizes your overall quality picture: total part records, total inspection runs, your rolling out-of-spec rate, and how many 8D investigations are still open. The Recent Inspection Runs table below shows the latest imports with an at-a-glance in-spec/flagged badge.



Dashboard view after adding two parts and importing four inspection runs.

Editing or Removing a Part

Use the **Edit** link in the Actions column to change a part's characteristics at any time — existing inspection runs keep their originally recorded values. **Delete** removes the part record itself; any historical runs already saved for it remain in your data but show as “Unlinked.”

PC-DMIS Routine & Vector Logic Auditor

This module scans PC-DMIS program text you paste in and flags common logic risks before they cause a bad part to ship: unconstrained alignment degrees of freedom, unstable 3-2-1 setups, flipped or corrupted hit vectors, and self-referencing DEPEND/VAR calculations.

Running an Audit

1 Get your routine text

In PC-DMIS, export or copy the program listing as plain text (File → Print to text, or select-all/copy from the edit window).

2 Paste or upload it

Paste directly into the text box, or click “Upload File” to load a .txt/.prg/.dmi file. Optionally link the audit to a part number from the dropdown for your records.

3 Click “Audit Routine”

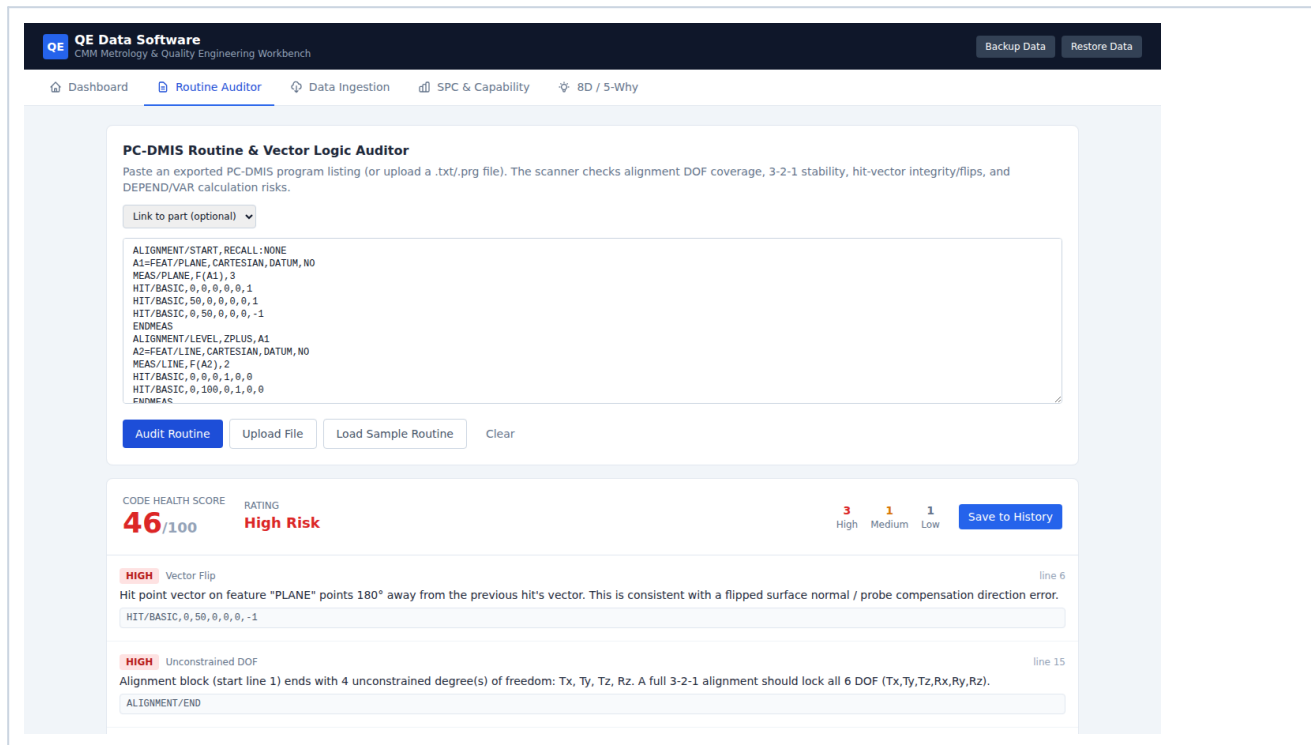
The scanner runs instantly and returns a 0–100 Code Health Score, a rating (Excellent / Good / Needs Attention / High Risk), and a line-by-line list of findings.

4 Review each finding

Findings are sorted High → Medium → Low severity and show the exact line number and offending code snippet. Use these as a checklist before releasing the program to the floor.

5 Save to History (optional)

Click “Save to History” to keep a permanent record of this audit, its score, and its findings for later reference or trending.



Audit results for a sample routine: a flipped hit vector and unconstrained rotation/translation DOF, both flagged High.

What the Scanner Checks

Unconstrained DOF. Confirms an ALIGNMENT block locks all 6 degrees of freedom (3 translation, 3 rotation) by the time it closes.

Unstable 3-2-1 Alignment. Flags a missing ALIGNMENT/LEVEL, ALIGNMENT/ROTATE, or ALIGNMENT/ORIGIN step, or a datum feature reused across multiple steps.

Vector Flip. Compares consecutive hit-point vectors on the same feature; a dot product suggesting >90° divergence is flagged as a likely flipped surface normal / probe-compensation error.

Vector Integrity. Flags hit vectors whose magnitude isn't close to 1.0 (a non-unit or corrupted vector).

Data Calculation. Flags DEPEND/VAR expressions that reference their own variable, or divide by a literal 0.

Housekeeping. Flags comments left over from debugging (TEMP, DEBUG, DO NOT USE, FIXME) so they don't ship by accident.

IMPORTANT

This is a heuristic, pattern-based scan of native PC-DMIS text — it is not a substitute for reviewing the alignment and vector logic against the CAD model and print. Treat a clean score as a first-pass sanity check, not a certification.

Inspection Data Ingestion & Normalization

Bring in raw CMM output — a PC-DMIS results CSV, or any spreadsheet export — and the app will compare every measurement to its part's tolerance and flag anything out of spec, before saving it as a permanent inspection run.

Importing a File

1 Open Data Ingestion

Drag and drop a .csv or .txt file onto the drop zone, click it to browse, or paste raw text directly into the box below it.

2 Click “Parse Data”

The app auto-detects your delimiter (comma, tab, or semicolon) and reads the header row to guess which column is which.

3 Check the column mapping

Confirm or correct the six dropdowns: Characteristic/Label and Actual are required; Nominal, Tol +, Tol -, and Serial Number are optional — if left blank, the app fills them in from the linked part's master tolerances.

4 Link to a part

Select the part number in “Link results to Part” so each row's Nominal/Tol are pulled from your master data, and so this run shows up in that part's SPC trend.

5 Review the live preview

Every row is evaluated instantly — out-of-spec rows are highlighted red with an OUT OF SPEC status. “Matched Char” shows Yes when the label matched a characteristic on the linked part.

6 Save the run

Click “Save Inspection Run.” It's now permanent and feeds the Dashboard, SPC charts, and the 8D anomaly picker.

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Backup Data

Restore Data

Dashboard

Routine Auditor

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8D / 5-Why

Column Mapping

Save Mapping Profile

Link results to Part (for tolerance lookup)

Serial / Unit ID (overrides mapped column if set)

10245-B (C)

SN-1001

Characteristic / Label *

Nominal

Actual (measured) *

Char

Nominal

Actual

Tolerance +

Tolerance -

Serial Number (optional)

Tol+

Tol-

(none)

Preview (2 rows)

All in spec

LABEL	NOMINAL	ACTUAL	TOL +/-	DEVIATION	STATUS	MATCHED CHAR
DIA_1	25	25.008	+0.02/-0.02	0.0080	OK	Yes
LENGTH_A	104.5	104.512	+0.05/-0.05	0.0120	OK	Yes

Save Inspection Run

Back

Column mapping and live preview after parsing a CSV — every row is checked against the linked part's tolerances instantly.

Reusable Mapping Profiles

If different machines or programmers export CSVs with different column orders, click **“Save Mapping Profile”** after setting up a mapping once, and give it a name. Next time you parse a file in that same format, pick it from the “Apply saved profile...” dropdown instead of remapping by hand.

TIP

The Label column is matched to your part's characteristic labels case-insensitively, with whitespace trimmed. If a row shows “Matched Char: No,” double-check the spelling matches the Dashboard exactly.

SPC Trend & Capability Visualizer

Once you have two or more saved runs for a characteristic, this module plots them over time and calculates process capability automatically — no spreadsheet required.

Viewing a Trend

1 Select a Part

Choose from the dropdown. Only parts with at least one saved run will show meaningful data.

2 Select a Characteristic

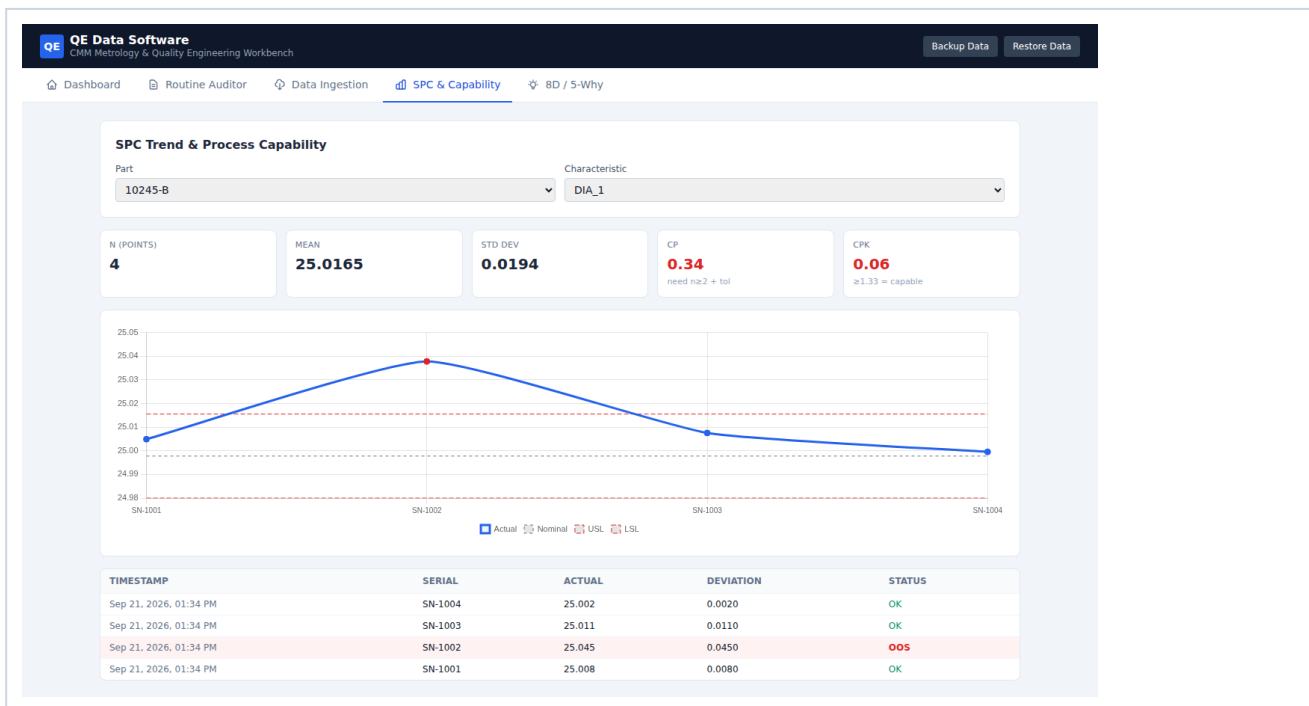
The dropdown lists every characteristic defined for that part on the Dashboard.

3 Read the stat tiles

n (points), Mean, Std Dev, Cp, and Cpk update instantly. Cp/Cpk require at least 2 data points and a defined nominal + tolerance.

4 Read the chart

Each point is one inspection run, plotted against dashed Nominal, USL, and LSL reference lines. Out-of-spec points are colored red.



SPC trend for DIA_1 across four runs — one out-of-spec point in red, with live Cp/Cpk and USL/LSL reference lines.

Interpreting Cp / Cpk

Cpk value	Meaning
≥ 1.33	Process is capable — shown in green.
1.00 – 1.32	Marginally capable — shown in amber; monitor closely.
< 1.00	Not capable — shown in red; expect out-of-spec parts.

TIP

Cp compares the tolerance band to your process spread; Cpk also accounts for how centered your process mean is within that band. A high Cp with a low Cpk means the process is precise but off-center — look at the Mean stat tile relative to the Nominal line on the chart.

8D & 5-Why Root Cause Assistant

When SPC or Data Ingestion flags an out-of-spec result, this module turns it into a formal 8D corrective action report with a multi-branch 5-Why root-cause chain — ready to print or share.

Starting an Investigation

- 1 Click “+ New Investigation”**
Opens a blank 8D form.
- 2 Pick a detected anomaly (optional)**
The “Start from a detected out-of-spec anomaly” dropdown lists every flagged result across all your saved runs. Selecting one auto-fills the Title and a D2 Problem Statement with the part, characteristic, serial, nominal, actual, and deviation.
- 3 Fill in D1–D3**
Enter the investigation team (D1), refine the problem statement (D2), and describe your immediate containment actions (D3) — e.g. segregating affected serials.
- 4 Build the 5-Why chain (D4)**
Answer “Why 1” through “Why 5” to drill from the symptom to the root cause. Click “+ Add Branch” to explore a second contributing cause path in parallel — give it a descriptive label like “Branch B: Fixturing.”
- 5 Complete D5–D8**
Document the corrective action plan, how it was implemented and validated, what prevents recurrence, and final closure/recognition.
- 6 Save the investigation**
Click “Save Investigation.” Update its Status (Open / In Progress / Closed) any time by reopening it for edits.

8D / 5-Why Investigation

Title: 10245-B - DIA_1 out of spec Status: Open

D1 - Team
Quality Engineer (lead), CMM Programmer, Production Supervisor

D2 - Problem Statement
On Sep 21, 2026, 01:34 PM, characteristic "DIA_1" on part 10245-B (Rev C), serial "SN-1002", measured 25.045 against nominal 25 (tol +0.02/-0.02). Deviation of 0.0450 exceeds the allowed tolerance band.

D3 - Containment Actions
Segregated affected lot (SN-1002 through SN-1010), placed on hold, 100% re-inspected DIA_1 on remaining stock.

D4 - Root Cause (Multi-Branch 5-Why) + Add Branch

Primary Remove branch

Why 1: DIA_1 measured 25.045 mm, exceeding the +0.020 mm upper tolerance.

Why 2: Probe deflected during approach due to excessive approach speed on a thin-wall section.

Why 3: Approach speed parameter was left at the default (rapid) value for this feature.

Why 4: Why did the previous cause happen?

Why 5: Why did the previous cause happen?

Cancel Save Investigation

The 8D form auto-filled from a detected anomaly, with a primary and a second 5-Why branch open.

Exporting a Report

From the investigation list, each row has two export links next to Edit and Delete:

- **Print** — opens a clean, formatted report in a new tab and triggers your browser's Print dialog, ready to Print or Save as PDF.
- **Markdown** — downloads a .md text file of the full report, ready to paste into an email, ticket, or another document.

8D / 5-Why Root Cause Investigations + New Investigation

Start from a detected out-of-spec anomaly or create a blank investigation.

TITLE	PART	STATUS	UPDATED	ACTIONS
10245-B - DIA_1 out of spec	10245-B	Open	Sep 21, 2026, 01:34 PM	Edit Print Markdown Delete

The investigation list — Edit, Print, Markdown, and Delete are available for every saved 8D.

8D Corrective Action Report

Part: 10245-B (Rev C) | Status: IN PROGRESS | Created: Sep 21, 2026, 01:32 PM | Updated: Sep 21, 2026, 01:32 PM

D1 - TEAM

Quality Engineer (lead), CMM Programmer, Production Supervisor

D2 - PROBLEM STATEMENT

On Sep 21, 2026, characteristic "DIA_1" on part 10245-B (Rev C), serial "SN-1002", measured 25.045 against nominal 25.000 (tol +0.02/-0.02). Deviation of 0.0450 exceeds the allowed tolerance band.

D3 - CONTAINMENT ACTIONS

Segregated affected lot (SN-1002 through SN-1010), placed on hold, 100% re-inspected DIA_1 on remaining stock.

D4 - ROOT CAUSE ANALYSIS (5-WHY)

Primary

1. **Why 1:** DIA_1 measured 25.045 mm, exceeding the +0.020 mm upper tolerance.
2. **Why 2:** Probe deflected during approach due to excessive approach speed on a thin-wall section.
3. **Why 3:** Approach speed parameter was left at the default (rapid) value for this feature.

Branch B: Fixturing

1. **Why 1:** Part was not fully seated against the tertiary locator before the cycle started.

D5 - CORRECTIVE ACTION PLAN

Reduce approach speed on DIA_1 in the CMM program; add a fixture seating sensor interlock.

D6 - IMPLEMENTATION & VALIDATION

Program updated and re-validated on 3 known-good parts; fixture interlock installed and verified on the line.

D7 - PREVENT RECURRENCE

Added approach-speed standard to the PC-DMIS programming checklist; updated operator training.

D8 - CLOSURE & TEAM RECOGNITION

Verified with Quality Manager on next production run; team recognized in weekly huddle.

The printable 8D report, generated from the same data, formatted for a clean PDF or hard copy.

IMPORTANT

If your browser blocks the Print report as a pop-up, allow pop-ups for this page (or this file) in your browser's address-bar icon, then click Print again.

Backing Up and Restoring Your Data

Because everything is stored locally in your browser, it's worth exporting a backup regularly — especially before clearing browser data, switching computers, or sharing records with a teammate.

Exporting a Backup

1 Click “Backup Data”

Found in the dark header bar on every screen. This downloads a single JSON file containing every part, run, audit, and 8D record.

2 Store it safely

Save it to a shared drive, USB stick, or backup folder. Name it with the date for easy version tracking.

Restoring a Backup

1 Click “Restore Data”

Choose the JSON backup file. You'll be asked to confirm before anything is loaded.

2 Confirm the merge

Restoring merges the backup into your current local database. Records with matching IDs are overwritten; everything else is added alongside what's already there.

GOOD TO KNOW

Use Backup Data / Restore Data to move your full working set from one computer to another — for example, from an office PC where you built out your part master to a shop-floor machine used for daily data ingestion.

Tips, Troubleshooting & FAQ

Q: The app looks unstyled or plain when I open it.

A: The page's visual styling loads from an internet CDN on first load. Connect to the internet once, or ask your IT team to allow `cdn.tailwindcss.com` if your shop-floor machine is on a restricted network. The SPC charts and all data entry still work without it.

Q: My CSV import shows “Matched Char: No” for every row.

A: The Label column values must match a characteristic Label on the linked part exactly (case-insensitive, whitespace trimmed). Check the Dashboard's Part Master for the exact spelling, or leave Nominal/Tol columns mapped directly from your file instead of relying on the part link.

Q: Cp/Cpk show as — (a dash) instead of a number.

A: You need at least 2 saved runs for that characteristic, and the part's Nominal, Tol +, and Tol - must all be filled in on the Dashboard.

Q: I don't see my new part in the Data Ingestion or SPC dropdowns.

A: Make sure you clicked “Save Part Record” in the editor — an unsaved draft won't appear elsewhere in the app.

Q: Can I use this on a tablet or shop-floor touchscreen?

A: Yes — the layout adapts to narrower screens for simple data-entry tasks like Data Ingestion, while denser views like SPC and the Part Master editor are best on a desktop-sized screen.

Q: Will I lose my data if I close the browser?

A: No. Everything is saved automatically to your browser's local database as you work. You will only lose data if you clear that browser's site data/history for this page, or open the app in a different browser or computer without restoring a backup first.

Q: How do I share records with a colleague?

A: Export a backup (Chapter 7) and send them the JSON file, or use each module's Print/Markdown export for a single report.

This guide reflects QE Data Software v1.0. For the underlying source files or to report a change you'd like made to the app itself, refer to the project repository your team maintains this tool in.